How Much Does Timing Matter? An Ablation of Explicit Duration Modelling in Text-to-Speech

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Advanced Machine Learning — Course Project
https://github.com/1Kunalvats9/audio_generation

Abstract—Neural text-to-speech systems learn phoneme durations and spectral content jointly, which makes it impossible to attribute output quality to either factor. We isolate the duration component by holding the acoustic model fixed and varying only the source of the durations. Using forced alignment on LJSpeech 1.1 we extract 949,856 ground-truth phone segments, train a classical duration predictor on 905,833 of them, and synthesise speech twice: once from measured durations and once from predicted ones. Gradient-boosted trees reduce duration error by 35.8 % relative to a per-phone-mean baseline, from 30.61 ms to 19.65 ms. However, substituting predicted durations for measured ones produced no measurable loss on either axis we evaluated: intelligibility (21.15 % vs 23.65 % word error rate; Wilcoxon $p = 0.048$, paired t-test $p = 0.100$) or subjective naturalness (mean opinion score 1.76 vs 1.84 over 25 listener-clip pairs, $p = 0.317$). By contrast, replacing real spectrograms with averaged phoneme templates cost 21.17 percentage points of word error rate and 2.32 MOS. We conclude that at this level of acoustic fidelity the acoustic model dominates duration accuracy by roughly a factor of thirty, and we report the alignment and significance diagnostics that support this claim.

Index Terms—text-to-speech, forced alignment, duration modelling, gradient boosting, ablation study, speech intelligibility

I. Introduction

A text-to-speech (TTS) system must solve two coupled sub-problems. It must decide how long each speech sound lasts, and it must decide what each sound looks like spectrally. The duration problem is non-trivial: in the word cat the vowel is substantially longer than either consonant, and the same phone lengthens before a pause and shortens in rapid speech.

Modern end-to-end systems learn both sub-problems inside a single network. This is effective but diagnostically opaque: when output quality is poor, the architecture offers no way to attribute the loss to timing rather than to spectral modelling. FastSpeech 2 [1] made duration explicit by replacing teacher-distilled durations with those produced by the Montreal Forced Aligner (MFA) [2], and routing them through a variance adaptor. The paper demonstrates that this improves quality overall, but does not isolate how much of that improvement is attributable to the duration predictor specifically.

This work performs that ablation. Our contributions are:

1) A fully reproducible classical TTS baseline on LJSpeech 1.1 in which the acoustic model is deliberately held fixed, so that duration is the only free variable.
2) A duration model with 21 text-derived features and an explicitly enforced no-leakage guarantee, evaluated against a non-trivial baseline on a source-disjoint test split.
3) A controlled comparison (A1 vs A2) quantifying the cost of predicting rather than measuring durations, together with the significance testing needed to interpret it.
4) The negative-result finding that this cost is statistically indistinguishable from zero, while the acoustic model accounts for essentially all degradation.

II. Related Work

A. Duration in neural TTS

FastSpeech 2 [1] is the direct antecedent of this work. Its central methodological change was to source phoneme durations from a forced aligner rather than from a teacher model, on the grounds that aligner durations are more accurate. Durations, pitch and energy are predicted at inference by the variance adaptor. Our experiment targets precisely the axis that paper introduced, and supplies the upper bound (A1, oracle durations) it does not report.

B. Forced alignment

The Montreal Forced Aligner [2] provides HMM/triphone alignment via Kaldi, producing phone-level boundaries from audio and its transcript. It is the standard source of ground-truth durations in the TTS literature and supplies our supervision signal. We treat its output as ground truth only after validating it (Section IV-A).

C. Neural vocoding

HiFi-GAN [3] reconstructs waveforms from mel spectrograms with high fidelity at low computational cost. We use a frozen pre-trained checkpoint so that the vocoder contributes an identical, constant transformation to every system under comparison.

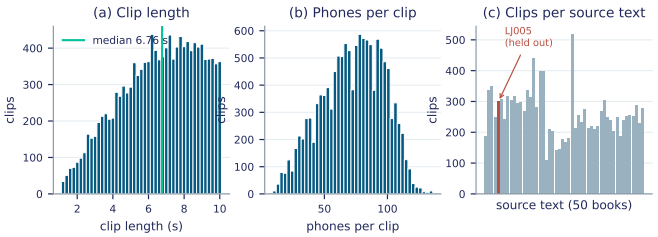


Fig. 1. Corpus statistics. (a) Clip length. (b) Phones per clip after alignment. (c) Clips per source text; the held-out book LJ005 is highlighted.

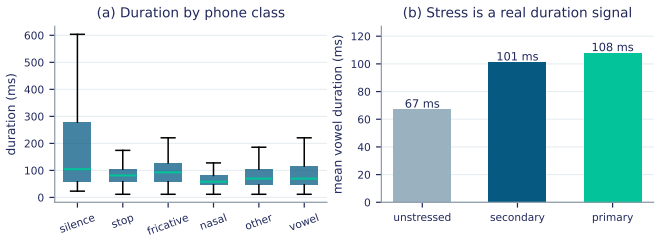


Fig. 2. Durations recovered by forced alignment. (a) By phone class. (b) The stress effect, which motivates retaining ARPAbet stress digits as distinct labels.

D. Diffusion-based speech generation

Grad-TTS [4] applies a score-based diffusion process in mel space, and AudioLDM [5] operates in a learned latent space. These represent the natural successor to the template-based acoustic model used here; Section VIII explains why our results point towards this component rather than towards better duration modelling.

E. Evaluation by automatic speech recognition

Human mean-opinion-score studies are the standard for naturalness but require a listening panel. Using an ASR system as a proxy listener gives a reproducible, automatic measure of intelligibility. We use Whisper [6] in its small.en configuration. We note explicitly that this measures intelligibility and not naturalness, a limitation revisited in Section VII-A.

III. Dataset and Exploratory Analysis

A. Corpus

LJSpeech 1.1 [7] comprises 13,100 utterances read by a single female speaker from seven public-domain non-fiction texts, sampled at 22,050 Hz, 16-bit mono, totalling 23.92 hours.

We verified the corpus rather than trusting its documentation. All 13,100 metadata rows have unique identifiers with exactly one matching waveform each (zero missing, zero extra); every file is 22,050 Hz mono. Clip length ranges from 1.11 s to 10.10 s with a median of 6.76 s, corresponding to 96–870 mel frames. Fig. 1 shows the distributions.

B. Text normalisation

The aligner requires orthographic text reducible to dictionary entries, so numerals and symbols must be expanded. Comparing the raw and supplied-normalised transcripts showed they differ in 1,505 rows (11.5 %), which determined that the normalised column is the correct source. After our cleaning, zero digits and zero non-ASCII characters remain across all 13,100 rows.

Three ambiguities were resolved by inspecting the data rather than by assumption. All 16 occurrences of St. are place names, with a single genuine Street case (LJ007-0230), so the many-case expansion to saint was adopted. The sole ft. occurrence is Dallas-Ft. Worth, giving fort. The 17 occurrences of messrs. were deliberately left unexpanded, since the MFA dictionary already contains messrs as M EH1 S ER0 Z, matching the speaker's realisation.

C. Partitioning

With a single speaker, speaker-disjoint evaluation is impossible. We therefore hold out whole source texts: all 299 clips of book LJ005 form the test set, guaranteeing that no test sentence shares its source document with any training sentence. This yields 12,501 training, 300 validation and 299 test clips; a fixed 100-clip subset (test_100) is used for synthesis and listening evaluation. All partitioning uses seed 42.

IV. Method

A. Forced alignment and its validation

MFA was run with the english_us_arpa acoustic model and dictionary. 366 words in LJSpeech are absent from that dictionary (proper nouns, compounds, acronyms such as FBI), affecting 1,179 clips; without pronunciations these clips would be discarded. We generated a supplementary dictionary so that all 13,100 clips aligned with none excluded, yielding 949,856 phone segments over 70 labels (39 ARPAbet phones with stress digits, plus silence).

Because every downstream result depends on these boundaries, we validated them with a test that includes its own control. At each speech–silence transition, frame energy should change in the expected direction. It does so for 94.6 % of 1,878 boundaries. Repeating the test with every boundary displaced by five frames (≈58 ms) gives 53.7 %, i.e. chance. A misaligned system would score equivalently on both; the separation is therefore evidence that the boundaries track real acoustic events.

As an independent check, the measured durations reproduce phonetic regularities we did not encode. DH is the shortest phone at 35 ms, diphthongs are longest, and stressed vowels average 108 ms against 67 ms for unstressed vowels (Fig. 2).

B. Acoustic representation

Audio is converted to 80-band log-mel spectrograms with a 1024-sample window and a 256-sample hop, giving

a frame period of $256/22050 = 11.61$ ms. All durations in this work are expressed in these frames.

The vocoder checkpoint was trained with a specific mel convention (amplitude rather than power spectra, Slaney normalisation and mel scale, natural logarithm after clamping at 10^{-5}). A mismatch here degrades audio in ways that are difficult to detect by ear, so we verified our implementation against the reference implementation numerically and obtained a mean absolute difference of 0.000000 before proceeding.

C. Feature engineering

The duration model predicts the length of each phone from 21 features, all derived from text alone. Five are categorical: the phone identity (retaining its stress digit), the preceding and following phone (with boundary tokens at clip edges, computed within clips so context never crosses an utterance), a six-way phone class, and the stress level. Sixteen are numeric: position within the word, word length, whether the phone is word-initial or word-final, position of the word within the utterance, position of the phone within the utterance, counts of phones and words, silence indicators for the phone and its neighbours, and the distance in phones to the nearest silence on each side.

The last pair encodes pre-pausal lengthening, the documented tendency of segments to lengthen before a pause. Fig. 3 confirms the model exploits it.

Leakage control. The most predictive available feature would be the utterance's true total length divided by its phone count — an excellent tempo estimate, and unavailable at synthesis time. We exclude every feature derived from true duration and enforce this with a unit test that multiplies all durations by three and asserts the resulting feature matrix is bit-identical. Notably this costs nothing: synthesis rescales predicted durations to the target utterance length, so any utterance-level tempo scalar is divided out again. The model need only recover durations relative to one another within an utterance.

D. Duration models

All models regress log(frames) rather than frames. Durations are strongly right-skewed — the median phone is 7 frames while silences reach 481 — so a squared-error objective on raw counts would allocate most capacity to a few long pauses. This matches the parameterisation used by FastSpeech 2's duration predictor.

Three models are compared, fitted on train, selected on val and reported once on test:

- Per-phone mean. The mean log-duration of each phone type, with a global fallback. A non-trivial baseline that any contextual model must beat.
- Ridge regression. One-hot encoded categoricals and standardised numerics; the penalty α was selected on validation from $\{0.1, 1, 10, 100\}$.
- Gradient-boosted trees. Histogram-based boosting with native categorical splits. Training halts on validation plateau, converging at 1,052 iterations.

TABLE I
The four systems. Each row changes one component relative to the row above.

System	Spectrogram	Durations
R0	— (original audio)	—
R1	real, from audio	real
A1	phoneme templates	measured (oracle)
A2	phoneme templates	predicted

E. Synthesis

To hold the acoustic model constant we use phoneme templates. For each of the 70 phones, every training occurrence is divided into three equal temporal states and averaged across occurrences, producing one fixed 80×3 matrix per phone. An utterance is synthesised by stretching each phone's three states to its assigned duration, concatenating, and passing the result to the frozen HiFi-GAN vocoder.

Predicted durations are emitted as unrounded reals and converted to integers that sum exactly to the target frame count, with every phone constrained to at least one frame.

V. Experimental Setup

Four systems are compared on the same 100 utterances, differing in exactly one component each:

The contrast of interest is A1 vs A2, which differ only in the source of the durations; templates, vocoder and target utterance length are identical. R0 establishes the error floor of the measuring instrument and R1 isolates the vocoder's contribution.

We evaluate on two axes. Intelligibility is word error rate (WER) and character error rate (CER) between the reference transcript and Whisper's hypothesis, with both strings passed through Whisper's EnglishTextNormalizer so that punctuation and orthographic variation are not scored. Naturalness is a mean opinion score (MOS): five listeners rated five utterances per system on a 1–5 scale, giving 25 ratings per system. The two measure different things — an utterance can be perfectly transcribable and still sound mechanical — so neither alone would support the conclusion.

VI. Results

A. Duration modelling

Table II reports performance on the 299 held-out utterances. Gradient-boosted trees reduce error by 35.8 % relative to the per-phone-mean baseline. The ordering (baseline > ridge > trees) is the expected one; an inversion would have indicated leakage or a defect. Validation and test figures agree closely for the selected model (1.702 vs 1.693 frames), indicating that selection did not overfit.

Every metric is reported twice, with and without silence. Silence constitutes only 5.3 % of segments but averages 14.6 frames against 7.4 for speech segments, with a maximum of 481; it therefore dominates an aggregate error

TABLE II
Duration prediction on the held-out book (299 utterances, 22,696 phones). One frame is 11.61 ms.

Model	MAE (frames)	MAE (ms)	MAE excl. sil. (ms)	RMSE (frames)
Per-phone mean	2.637	30.61	26.88	4.34
Ridge	2.338	27.14	23.31	3.93
Boosted trees	1.693	19.65	16.84	3.08

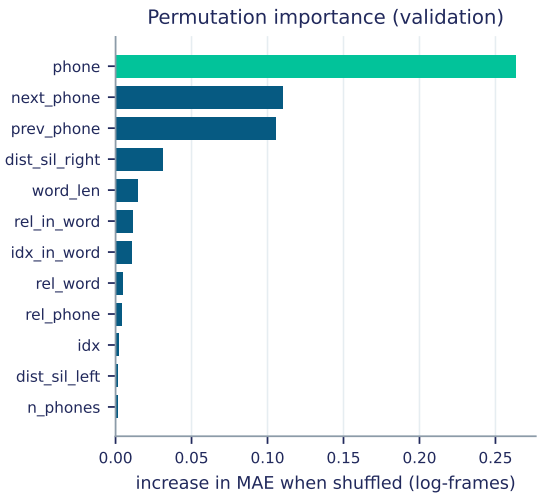


Fig. 3. Permutation importance of the twelve strongest features, measured as the increase in validation MAE when a feature is shuffled.

figure. The silence-excluded column is the more faithful measure of learned phone duration, while the inclusive column is what synthesis actually consumes.

Fig. 3 gives permutation importance on the validation split. Phone identity dominates, as expected, but the contextual features contribute materially — in particular the distance-to-silence pair, confirming that the model recovers pre-pausal lengthening from data.

B. Intelligibility

Table III reports WER and CER for all four systems.

Decomposing the chain attributes the degradation to specific components:

- R0 → R1: +0.63 pp. The mel representation and the vocoder are nearly lossless. R0's own non-zero floor is partly an artefact of reference normalisation: three of its ten worst utterances are cases where the reference reads "george the 4th" and Whisper outputs "george iv".
- R1 → A1: +21.17 pp. Replacing real spectrograms with averaged templates accounts for essentially all loss — roughly 34 times the vocoder's cost.
- A1 → A2: −2.50 pp. Predicting durations rather than measuring them did not degrade intelligibility.

TABLE III
Intelligibility, 100 utterances per system, Whisper small.en.

System	Description	WER	CER
R0	original recording	1.85 %	0.71 %
R1	real mel → vocoder	2.48 %	0.95 %
A1	templates + measured durations	23.65 %	11.83 %
A2	templates + predicted durations	21.15 %	10.02 %

TABLE IV
Subjective naturalness. Five listeners × five utterances per system.

System	Description	MOS	SD
R0	original recording	4.88	0.33
R1	real mel → vocoder	4.16	0.37
A1	templates + measured durations	1.84	0.37
A2	templates + predicted durations	1.76	0.44

C. Statistical analysis of the A1–A2 contrast

The sign of the A1–A2 difference is opposite to our expectation, so we tested it rather than reporting it at face value. On the same 100 utterances, A2 is better on 40, worse on 30 and tied on 30. A Wilcoxon signed-rank test gives $p = 0.048$; a paired t-test gives $p = 0.100$.

One test crosses the conventional threshold and the other does not. We therefore do not claim that predicted durations are superior to measured ones. The defensible conclusion is the weaker and more interesting one:

> At this level of acoustic fidelity, duration prediction error of 19.65 ms has no measurable effect on intelligibility. The acoustic model, not the duration model, is the binding constraint.

D. Subjective naturalness

Table IV reports the listening study. It reproduces the ordering of the WER results and sharpens the attribution. The vocoder costs 0.72 MOS, whereas the averaged templates cost 2.32 MOS — again the dominant term. Between A1 and A2, 21 of 25 listener-clip pairs are tied, 3 favour A1 and 1 favours A2 ($p = 0.317$), so the duration source has no significant effect on naturalness either. Per-listener means lie between 3.05 and 3.30, indicating no outlying rater.

This is the stronger form of our main claim: the duration source is not detectable by either instrument, while the acoustic model is obvious to both.

E. Why is A2 not worse? A partial explanation

We investigated the mechanism and report an unconfirmed hypothesis rather than a conclusion.

Our first conjecture — that regression toward the mean smooths the predictions — is false: predicted durations are more dispersed than measured ones ($\sigma = 7.04$ vs 5.01 frames), partly because the rescaling step assigns residual frames to the longest segment, producing phones of up to 154 frames against a true maximum of 76.

The surviving hypothesis concerns very short segments. A three-state template cannot be rendered in a single frame. Measured durations place 3.5 % of segments at two frames or fewer, whereas predicted durations place only 0.8 % there. Across utterances, the more short segments A2 eliminated, the more it improved ($r = +0.227$, $p = 0.023$). However, A1's own error does not track its short-segment fraction ($r = +0.014$, $p = 0.89$), so the mechanism is not established; we report it as a direction for further work.

VII. Discussion

A. Limitations

A2 receives oracle utterance length. Predicted durations are rescaled to the true total length, information a deployed system would not possess. Our comparison therefore isolates the relative distribution of durations, not total-length prediction, and A2 is not a fully autonomous system. We adopted this to ensure both systems present the vocoder with identically-sized inputs.

The listening study is small and not blind. Twenty-five ratings per system from five listeners, who are the authors, is enough to establish the large R1–A1 gap but not to resolve small differences. The raters knew the systems were being compared, though not which file was which, so the MOS figures should be read as corroboration of the WER ordering rather than as an independent measurement of naturalness.

A single speaker. Speaker generalisation cannot be assessed; the source-disjoint split is the strongest available substitute.

Deliberately weak acoustic model. Phoneme templates discard coarticulation and all prosody beyond duration. This is required by the experimental design but bounds the attainable quality of A1 and A2.

B. Implications for FastSpeech 2

Our results support the premise that forced-aligner durations are accurate, but suggest that under a weak acoustic model, improvements to the duration predictor yield little gain. We initially supposed the variance adaptor's value would show up in naturalness, where WER is blind. The listening study of Section VI-D does not bear that out: A1 and A2 are statistically indistinguishable on MOS as well. The more likely reading is that an acoustic model this coarse masks duration effects entirely, and that the variance adaptor's contribution would only become visible once the acoustic model is strong enough for prosody to be perceptible at all.

C. Engineering notes

The repository carries 170 automated tests, including tests that guard shared numerical contracts: the frame-count formula, the mel convention against the vocoder's reference implementation, and the no-leakage property of Section IV-C. Three substantive defects were found only by executing the full pipeline rather than by inspection: a residual-distribution error that corrupted 27 of 299 utterances, a type error that aborted synthesis, and an empty-slice NaN that silently invalidated 61 of 70 templates. Each is now covered by a regression test.

VIII. Conclusion and Future Work

We isolated the duration component of a TTS pipeline by holding the acoustic model fixed. Gradient-boosted trees on 21 text-derived features reduced duration error by 35.8 % over a per-phone-mean baseline. Substituting those predictions for oracle durations produced no measurable loss in either intelligibility or subjective naturalness, while the averaged-template acoustic model accounted for 21.17 percentage points of word error rate and 2.32 MOS.

The practical implication is that further effort belongs in the acoustic model rather than in duration modelling. Diffusion-based generators [4], [5] are the natural replacement, and the alignment and duration infrastructure developed here supplies exactly the conditioning signal such a model requires. Three concrete next steps follow: predict total utterance length so that A2 becomes autonomous; distribute the rescaling residual proportionally rather than onto a single segment; and repeat the listening study blind, with naive listeners and more utterances, once a stronger acoustic model makes prosodic differences perceptible.

Contributions

Kunal Vats — corpus preparation, text normalisation, partitioning, shared configuration; duration model (with Pushkar Jain). Pushkar Jain — duration model: feature construction, model comparison, error analysis (with Kunal Vats). Tanishk Agrawal — forced alignment, out-of-vocabulary dictionary, alignment validation. Shaurya Sharma — mel extraction, phoneme templates, vocoder integration, synthesis. Chaitanya Shekhawat — ASR evaluation harness, results aggregation, failure analysis.

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